4.4

PAEDIATRICS

Ophthalmology Infections

CHAPTER 4.4	OPHTHALMOLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
Keratitis			
Older Child *S aureus* *Moraxella catarrhalis* *Streptococcus pneumoniae* *Pseudomonas aeruginosa* *Klebsiella pneumoniae*	**Topical eye drops:** Chloramphenicol Fusidic Acid Gentamicin	**Topical eye drops:** Ciprofloxacin 0.3% Polymyxin B	• Choice depends on causative organisms and clinical suspicions. • Pseudomonas aeruginosa keratitis is usually associated with contact lens use. • *Not available in UMMC.
Fungal: *Candida* spp (most common) *Fusarium* spp *Aspergillus* spp	**Topical antifungals:** Fluconazole 1% Miconazole 1%	**Topical antifungals:** Amphotericin B 0.15-1%	• Use in conjunction with systemic antifungal.
Herpes simplex Virus	Topical Aciclovir five times daily for 10 - 14		

CHAPTER 4.4	OPHTHALMOLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
	day; continue ≥ 3 days post-healing + Aciclovir PO: **1 - 24 months old:** 100 mg **2 - 18 years old:** 200 mg 5 times daily for 7 - 14 days		
Varicella zoster Virus	Topical antimicrobials to prevent secondary bacterial infection + Aciclovir PO: **1-23 months:** 200 mg q6H for 5 days **2-5 years:** 400 mg q6H for 5 days **6-11 years:** 800 mg q6H for 5 days		• Severe cases may benefit from steroid eye drops.

CHAPTER 4.4	OPHTHALMOLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
	12-17 years: 800 mg 5 times daily for 7 days		
Conjunctivitis			
Bacterial: *Haemophilus influenzae* *Moraxella catarrhalis* *S pneumoniae* *Staph aureus* *S viridians* Pseudomonas	Chloramphenicol eye drops q4 - 6H during daytime and eye Ointment at night **OR** Gentamicin eye drops **OR** Fusidic ointment	Tetracycline eye drops during the day and ointment once during night.	• Ciprofloxacin or Norfloxacin seems to be the widest spectrum for bacterial conjunctivitis.
Chlamydia trachomatis	Ciprofloxacin 0.3% **OR** Norfloxacin Eye Drops		• Erythromycin / Gentamicin eye drops for Chlamydia trachomatis conjunctivitis in neonates. • Systemic antibiotics may be needed when associated with otitis media.
Viral: Adenoviruses *Varicella zoster virus*	**Local infection:** No suitable topical antibiotic		• Immunocompromised patients with advanced infection may need systemic Cidofovir.

CHAPTER 4.4	OPHTHALMOLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
	Topical Acyclovir for VZV		
Endophthalmitis			
Post-Traumatic / Operative **Acute (within 48 - 72 hours):** CONS *S aureus* Streptococci *Pseudomonas* spp Bacillus spp Candida (usually *C. parapsilosis*) Aspergillus (rare) **Delayed:** CONS *Propionibacterium acnes* Candida Aspergillus **Endogenous**	**Intravitreal Antimicrobials:** Ceftazidime IVT + Vancomycin IVT	**Intravitreal Antimicrobials:** Vancomycin IVT + Aminoglycosides (Gentamicin **OR** Amikacin) IVT	• Intravitreal injection of Gentamicin may cause retinal injury. • Ultimate choice of antimicrobial depends on smear taken at operation for microscopy, stain, C&S. • Systemic antibiotics, depending on the organism suspected or isolated – Vancomycin IV, Ceftriaxone IV. • Early starting of antifungal has better prognosis.

CHAPTER 4.4	OPHTHALMOLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
(Haematogenous): Streptococci Staphylococci Candida *Bacillus* spp Enteric Gram-negative bacilli			
Preseptal Cellulitis			
Bacterial: *Streptococci* *S aureus* *Haemophilus influenzae b*	Co-amoxiclav 25 mg/kg/dose (Amoxicillin component) IV q8H **OR** Cloxacillin 50 mg/kg/dose IV q6H **Systemically unwell:** Ceftriaxone 50 mg/kg/dose IV q12H	Cefuroxime 50 mg/kg/dose IV q8H	• Associated with entry site lesion on skin: • MSSA: Cloxacillin 50mg/kg/dose IV q6H • CA-MRSA: Clindamycin 10mg/kg/dose IV q8H
Orbital Cellulitis			

CHAPTER 4.4	OPHTHALMOLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
Bacterial: *Streptococci* *S aureus* *Haemophilus influenzae b* *Pseudomonas* *Klebsiella* *Enterococcus*	Ceftriaxone 50 mg/kg/ dose IV q12H + Cloxacillin 50 mg/kg/ dose IV q6H **Urgent CT scan is needed to exclude abscess and intracranial extension. If evidence of abscess, ADD:** Metronidazole 7.5 mg/kg/dose IV q8H	Ceftriaxone 50 mg/kg/ dose IV q24H + Clindamycin 10 mg/kg/ dose IV q8H	• Duration: 2 weeks • **Penicillin allergy, CA-MRSA:** Use Clindamycin. • Final choice of antibiotic is based on culture and sensitivity.
Fungal: Mucormycosis *Aspergillus* spp	Amphotericin B Deoxycholate 1-1.5 mg/kg IV q24H	**Patient > 2 years old:** Voriconazole 9 mg/kg IV q12H for 2 doses on Day 1, **then** maintenance 8 mg/kg IV q12H. **For older children > 13 years old:**	• Fungal orbital/facial cellulitis more often in immunocompromised. • If tubulopathy or nephrotoxicity, use Liposomal Amphotericin B 3-5 mg/kg IV q24H • Certain Aspergillus species may be sensitive to the Echinocandin group.

CHAPTER 4.4	OPHTHALMOLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
		Consider Posaconazole	
Retinitis			
Cytomegalovirus (CMV)	**Intravitreal:** Ganciclovir 2.5 mg IVT **Initial Therapy:** Ganciclovir 5 mg/kg/ dose IV q12H for 2-3 weeks **Maintenance:** Valganciclovir 16 mg/kg/ dose PO q24H for 3 weeks	**Initial Therapy:** Foscarnet* 60 mg/kg/ dose IV q8H for 2-3 weeks, **then** 90 - 120 mg/kg/dose q24H for chronic suppressive therapy	• Repeat induction regimen if CMV retinitis progresses while on therapy. • * Foscarnet is not available in UMMC formulary.
Herpes simplex Chorioretinitis	Aciclovir 10 mg/kg/dose IV q8H for 5 - 10 days **Followed by** Aciclovir PO:		

CHAPTER 4.4	OPHTHALMOLOGY INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
	1 - 24 months old: 100 mg **2 - 18 years old:** 200 mg 5 times daily for 4-6 weeks		
Toxoplasma gondii	Pyrimethamine PO: 2 mg/kg/dose q24H for 2 days; then, 1 mg/kg/dose q24H for 6 months; then, 1 mg/kg/day 3 times per week for 6 months (max: 25 mg/dose) + Sulfadiazine 50 mg/kg/dose PO q12H for 12 months		• Duration: 12 months. • Add Folinic Acid 10 - 25 mg PO daily while on treatment and continue for 1 week after discontinuation.